

Prerequisite

You'll need a Kubernetes (K8s) cluster for the homework.

You can use the iac cluster but delete it at the end of the day

Provide the mentors with manifests or commands for your homeworks.

Exercise 1

Starting from the Pod exercise from yesterday do the following:

- Create a Deployment with a replica of 2
- Configure the Pod to require at least *200 Mb* and *0.2 CPU*
- Configure the Pod to not use more than *0.5 CPU*
- `default` namespace must be used here

Exercise 2

Use the *Secret* and *ConfigMap* to configure the above deployment with the following:

- Use a custom html file for the nginx pods
- For the current nginx image, create a copy of it and tag it for your personal private repository (DockerHub); replace the image with the one from your private repository

Exercise 3

Create a *Service* of type *NodePort* for the above deployment. The port used should be `8080`.

Exercise 4

Create a Deployment `echo` with the following specifications:

- image used should be <https://hub.docker.com/r/ealen/echo-server>
- deployment should be in the `playground` namespace

Create a *Service* with ClusterIP for the above service. Check documentation for port.

Exercise 5

Create a NetworkPolicy that denies all incoming requests in the `playground` namespace and allows only the `nginx` from the `default` namespace.

- check if you can do a curl from the `nginx` to `echo`

Exercise 6

Create a Job that:

- runs a curl command against the `echo` service
- if the curl succeeds output the following `Network Policy failed! I shouldn't be allowed!`
- if the curl fails output the following `Network Policy works! Echo is safe <3`

Extra Exercise

Manually (UI) create a LB on Azure that:

- forwards requests to the worker nodes of K8s
- update the nginx configuration so that the requests are getting forwarded to the `echo` service
- the response of the request should come from the `echo`